

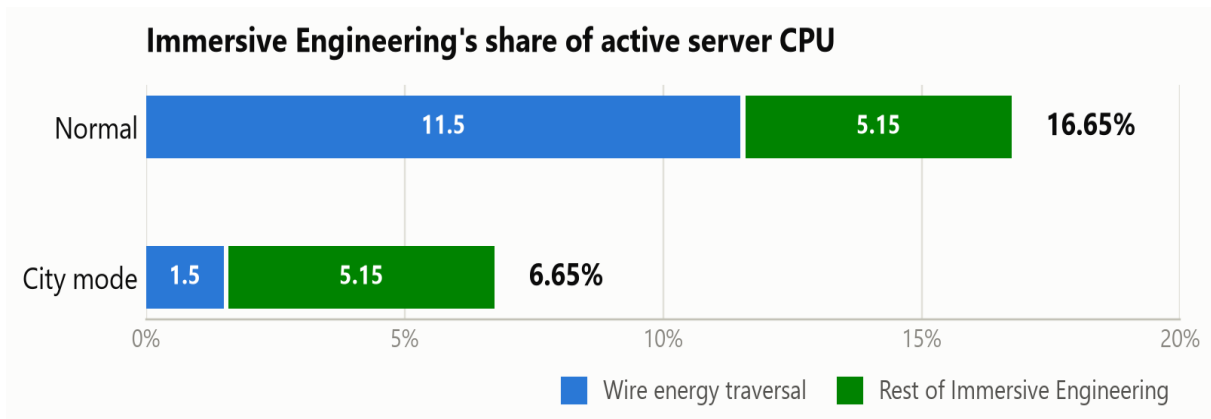
City Mode

A config-gated power simulation for Immersive Engineering 1.12.2 that keeps the wiring and removes the grid maths.

City mode keeps every visible part of the electrical system — connectors, relays, transformers, breaker switches, energy meters and the catenary wires between them — while replacing the simulation behind them with a single lossless push. It exists because on a large build the realistic grid is the most expensive thing this mod does per tick.

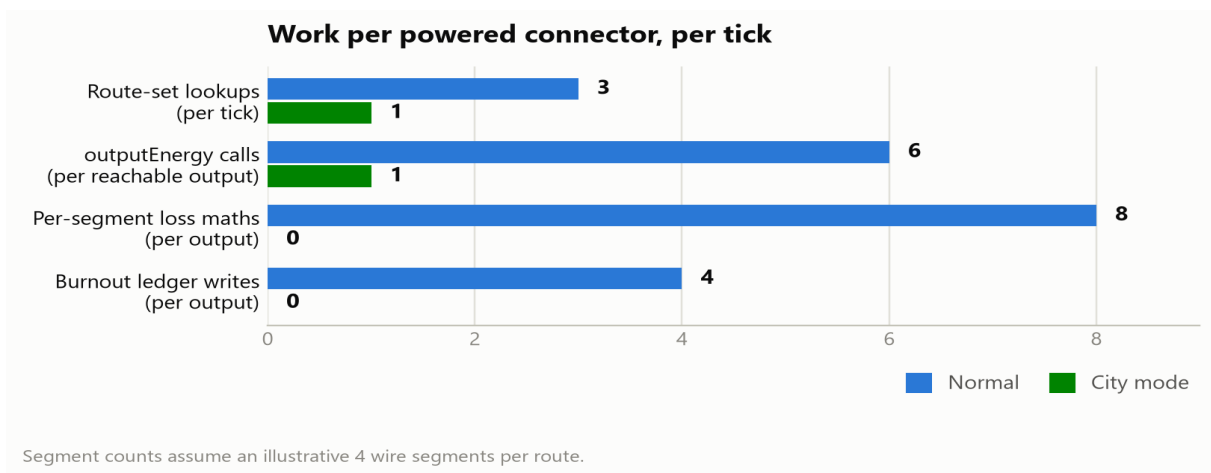
It is **off by default**. With `cityMode = false` none of the paths described here are entered and behaviour is byte-identical to stock.

Modelled performance



These are modelled figures, not measurements. They apply the operation counts below to the one measured baseline this fork has — a live spark profile of the production server, where Immersive Engineering was the single most expensive individual mod at 16.65% of active CPU, roughly 11.5 points of which was the wire traversal alone. City mode itself has not yet been profiled under live load. Treat the shape as sound and the exact percentages as an estimate.

Where the saving comes from



The reduction is structural rather than incremental. Normal mode fetches the route set three times per tick and runs the distribution twice — a simulate pass to discover demand, then a real pass that sorts every output into a TreeMap by loss rate, gives each a proportional share, applies per-segment attenuation and records throughput for wire burnout — followed by a third full walk of the network to advertise available energy. City mode does one lookup, one real push per output, and stops.

What changes, in gameplay terms

	Normal	City mode
Wire loss	2.5–5% per 16 blocks, worse when lightly loaded	none
Voltage tiers	throttle to the weakest wire on the path	cosmetic only
Supply < demand	proportional to demand, nearest first	greedy, arbitrary order
Wire burnout	wire destroyed above its rate	cannot happen
Wire shock damage	sourced from the whole network	sourced from the local connector
Breaker switches	cut the network	unchanged
Energy meters	read loss-attenuated throughput	read full throughput
Generator fuel	consumed only under load	unchanged
Machine power needs	—	unchanged

Two consequences worth knowing

Wire burnout is disabled

Overload destruction works by the normal transfer path recording per-connection throughput into a ledger that a world-tick handler then acts on. That path is the ledger's only writer, so in city mode it stays empty and no wire can ever burn out. Combined with the fact that city mode never clamps to the cable's rate, a copper wire will carry a full HV connector's 4096 FE/t forever with no consequence. That is intended for a city pack, but it is a rules change rather than an optimisation.

Wire shock damage becomes local

Damage is computed from a per-tick list of energy sources on each connectable, which the skipped network broadcast used to populate. In city mode that list holds only the connector's own energy, so damage is generally lower and a wire span running between two relays will not shock at all, because relays never hold energy. Set `enableWireDamage = false` to turn it off properly.

Does a diesel generator still need fuel?

Yes. No generator reads the city-mode flag — the entire footprint is four reads of one boolean inside the connector class. A diesel generator gates fuel consumption on two conditions city mode never touches: something must actually accept power, and there must be registered fuel in the tank. With no consumers, connector buffers saturate, the generator's simulated insert returns zero, the fan spins down and no fuel burns — in both modes.

What city mode changes is the yield per unit of fuel: the 4096 FE/t leaving the generator arrives undiminished instead of being attenuated by every wire segment on the way. More useful power per bucket of biodiesel, but never power from nothing — only what receivers actually accepted is deducted from the connector.

Block-by-block summary

Block	Role	Affected by city mode
Diesel Generator	4096 FE/t, burns fuel only under load	No — but its output now arrives lossless
Thermoelectric / Dynamo	push to adjacent blocks only	No
Windmill / Water Wheel	drive the dynamo, produce no FE	No
Capacitor LV/MV/HV	buffer; 256 / 1024 / 4096 FE/t	No
Creative Capacitor	infinite source, all six sides	No — works as an infinite source in both modes
Lightning Rod	16,000,000 FE on a strike	No
Connector LV/MV/HV	the only blocks moving energy over wires	Yes — push path replaced; its own rate caps still apply
Relay LV/MV/HV	routing waypoint, never holds energy	No — tick body never runs either way
Transformer	tier adapter; holds no energy	Yes — tier throttling disappears
Breaker Switch	interrupts the network	No — still cuts
Energy Meter	passive throughput accumulator	Works, and reads the full unattenuated amount
Machines	receive by push into their own buffer	No — their own intake caps still apply
Direct generator → machine	no wires involved	No — entirely unaffected

Full technical detail, including the annotated transfer routine, wire-type tables and the testing checklist, is in `docs/CITY_MODE.md`.